

Week #1 Checkpoint

ducklake_snapshots('lake'):

snapshot_id int64	snapshot_time timestamp with time zone	schema_version int64	...	author varchar	commit_message varchar	commit_extra_info varchar
0	2026-06-27 03:40:38.110246+00	0	...	NULL	NULL	NULL
1	2026-06-27 03:40:38.157545+00	1	...	NULL	NULL	NULL
2	2026-06-27 03:40:38.169796+00	2	...	NULL	NULL	NULL
3	2026-06-27 03:40:38.196537+00	3	...	NULL	NULL	NULL
4	2026-06-27 03:40:38.222061+00	4	...	NULL	NULL	NULL
5	2026-06-27 05:46:48.92686+00	5	...	NULL	NULL	NULL
6	2026-06-27 05:46:49.000416+00	6	...	NULL	NULL	NULL

7 rows

7 columns (6 shown)

RustFS raw objects (sample):

```
raw/coco/images/000000000192.jpg (230425 bytes)
raw/coco/images/000000000283.jpg (146834 bytes)
raw/coco/images/000000000370.jpg (84389 bytes)
raw/coco/images/000000000459.jpg (194898 bytes)
raw/coco/images/000000000474.jpg (131485 bytes)
raw/visdrone/00000001/00000001-2.jpg (224567 bytes)
raw/visdrone/00000001/00000001-3.jpg (437841 bytes)
raw/visdrone/00000001/00000001-4.jpg (196739 bytes)
raw/visdrone/00000001/00000001-5.jpg (476081 bytes)
raw/visdrone/00000001/00000001-6.jpg (267720 bytes)
```

raw.coco_annotations sample URIs:

image_id int64	file_name varchar	uri varchar
474	000000000474.jpg	s3://lakehouse/raw/coco/images/000000000474.jpg
2529	000000002529.jpg	s3://lakehouse/raw/coco/images/000000002529.jpg
3145	000000003145.jpg	s3://lakehouse/raw/coco/images/000000003145.jpg
3735	000000003735.jpg	s3://lakehouse/raw/coco/images/000000003735.jpg
589	000000000589.jpg	s3://lakehouse/raw/coco/images/000000000589.jpg

Versioning Demo / Week #2 Checkpoint

=== Snapshot history so far (DuckLake's own change log doubles as our notes) ===

snapshot_id int64	snapshot_time timestamp with time zone	schema_version int64	changes map(varchar, varchar[])
0	2026-06-27 06:31:29.730795+00	0	{schemas_created=[main]}
1	2026-06-27 06:31:29.787821+00	1	{schemas_created=[raw]}
2	2026-06-27 06:31:29.80201+00	2	{schemas_created=[silver]}
3	2026-06-27 06:31:29.827509+00	3	{schemas_created=[gold]}
4	2026-06-27 06:31:29.852776+00	4	{tables_created=[raw.coco_annotations], tables_inser...
5	2026-06-27 06:31:29.922096+00	5	{tables_created=[raw.visdrone_fragments], tables_ins...
6	2026-06-27 06:31:29.991936+00	6	{tables_created=[raw.coco_categories], tables_insert...
7	2026-06-27 06:31:53.719974+00	7	{tables_created=[silver.coco_annotations], tables_in...
8	2026-06-27 06:31:53.813367+00	8	{tables_altered=[7]}
9	2026-06-27 06:31:53.868261+00	9	{tables_created=[silver.visdrone_fragments], tables_...
10	2026-06-27 06:31:53.961698+00	10	{tables_created=[gold.coco_labels], tables_inserted_...
11	2026-06-27 06:31:54.051053+00	11	{tables_created=[gold.visdrone_training], tables_ins...

12 rows

4 columns

=== Time travel by VERSION: schema before/after the is_crowd rename ===

schema AT (VERSION => 7) (before rename):

column_name varchar	column_type varchar	null varchar	key varchar	default varchar	extra varchar
image_id	BIGINT	YES	NULL	NULL	NULL
file_name	VARCHAR	YES	NULL	NULL	NULL
uri	VARCHAR	YES	NULL	NULL	NULL
width	INTEGER	YES	NULL	NULL	NULL
height	INTEGER	YES	NULL	NULL	NULL
annotation_id	BIGINT	YES	NULL	NULL	NULL
category_id	BIGINT	YES	NULL	NULL	NULL
bbox	DOUBLE[]	YES	NULL	NULL	NULL
area	DOUBLE	YES	NULL	NULL	NULL
is_crowd	BOOLEAN	YES	NULL	NULL	NULL

10 rows

6 columns

schema AT (VERSION => 8) (after rename):

column_name varchar	column_type varchar	null varchar	key varchar	default varchar	extra varchar
image_id	BIGINT	YES	NULL	NULL	NULL
file_name	VARCHAR	YES	NULL	NULL	NULL
uri	VARCHAR	YES	NULL	NULL	NULL
width	INTEGER	YES	NULL	NULL	NULL
height	INTEGER	YES	NULL	NULL	NULL
annotation_id	BIGINT	YES	NULL	NULL	NULL
category_id	BIGINT	YES	NULL	NULL	NULL
bbox	DOUBLE[]	YES	NULL	NULL	NULL
area	DOUBLE	YES	NULL	NULL	NULL
is_crowd_annotation	BOOLEAN	YES	NULL	NULL	NULL

10 rows

6 columns

=== Time travel by TIMESTAMP ===
row count AT (TIMESTAMP => '2026-06-27 06:31:53.719974+00:00'):

count_star() int64
117

=== Exercising version control on silver.coco_annotations ===
Good snapshot 11: 40 distinct category_id values

Running a deliberately bad transform (wipes every category_id)...
Bad snapshot 12: 1 distinct category_id value(s) (corrupted)

=== Comparing snapshot 11 vs 12 via ducklake_table_changes ===

change_type varchar	row_count int64
update_postimage	117
update_preimage	117

Rolling back: restoring data from snapshot 11...
Restored snapshot 13: 40 distinct category_id values

Bad snapshot 12 is still time-travel queryable - history is immutable, rollback only adds a new corrective snapshot:

still_corrupted_in_history int64
1

Video Fragment Query / Week #2 Checkpoint

=== New snapshots from this run ===

snapshot_id int64	schema_version int64	changes map(varchar, varchar[])
11	11	{tables_created=[gold.visdrone_training], tables_inserted_into=[10]}
10	10	{tables_created=[gold.coco_labels], tables_inserted_into=[9]}
9	9	{tables_created=[silver.visdrone_fragments], tables_inserted_into=[8]}
8	8	{tables_altered=[7]}
7	7	{tables_created=[silver.coco_annotations], tables_inserted_into=[7]}
6	6	{tables_created=[raw.coco_categories], tables_inserted_into=[6]}
5	5	{tables_created=[raw.visdrone_fragments], tables_inserted_into=[5]}
4	4	{tables_created=[raw.coco_annotations], tables_inserted_into=[4]}

=== COCO metadata query: crowded scenes (iscrowd flag) ===

image_id int64	file_name varchar	object_count int64	labels varchar[]
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0 rows

=== COCO metadata query: crowded scenes (iscrowd flag) ===

image_id int64	file_name varchar	object_count int64	labels varchar[]
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0 rows

=== COCO metadata query: most object-dense scenes (fallback view) ===

image_id int64	file_name varchar	object_count int64	primary_label varchar	split varchar
3145	000000003145.jpg	21	chair	val
2477	000000002477.jpg	10	person	train
1915	000000001915.jpg	8	person	val
3938	000000003938.jpg	8	person	train
1290	000000001290.jpg	7	cake	val

=== VisDrone fragment query: select keyframes only by per-fragment stats ===

video_id varchar	fragment_index int32	uri varchar	byte_size int64
0000001	1	s3://lakehouse/raw/visdrone/0000001/0000001.jpg	165707
0000002	1	s3://lakehouse/raw/visdrone/0000002/0000002.jpg	166304
0000003	1	s3://lakehouse/raw/visdrone/0000003/0000003.jpg	166324

Fetches 3 keyframe object(s), 498335 bytes total.

Those same 3 video(s) have 21 frames totaling 9035790 bytes.

-> read 5.5% of the clip bytes: confirms only the selected fragments were read from RustFS, not whole clips.

Week #3 Huggin Face Round Trip

<https://huggingface.co/datasets/Acender/cs375-coco-gold>
<https://huggingface.co/datasets/Acender/cs375-visdrone-gold>

```
[COCO incremental] looking for 10 new images from aegean-ai/coco-25k
uploaded 10 new image blobs to s3://lakehouse/raw/coco/images/
Using cached annotations at /data/local/coco_annotations_src/instances_train2017.json
staged 90 new annotation rows at /data/local/coco_annotations_increment.parquet (local-storage staging area)
inserted into raw.coco_annotations -> lakehouse
[VisDrone incremental] looking for 1 new sequence(s) from Voxe151/visdrone-mot
uploaded 7 new frame blobs across 1 sequence(s)
staged 7 new fragment rows at /data/local/visdrone_fragments_increment.parquet (local-storage staging area)
inserted into raw.visdrone_fragments -> lakehouse
```

=== New snapshot(s) from this incremental load ===

snapshot_id int64	schema_version int64	changes map(varchar, varchar[])
14	12	{tables_inserted_into=[4]}
15	12	{inlined_insert=[5]}

Show Populated Lakehouse

0000004	2	7	s3://lakehouse/raw/v...	...	false	0.2857142857142857	train
0000004	3	7	s3://lakehouse/raw/v...	...	false	0.42857142857142855	train

3 rows

8 columns (7 shown)

=== RustFS object counts (the actual image/video bytes) ===

s3://lakehouse/raw/coco/images/ -> 35 object(s), 5,201,376 bytes

s3://lakehouse/raw/visdrone/ -> 28 object(s), 12,058,473 bytes

=== Cross-check: catalog URIs actually resolve to RustFS objects ===

OK s3://lakehouse/raw/coco/images/000000000192.jpg (230425 bytes, image/jpeg)

OK s3://lakehouse/raw/coco/images/000000000192.jpg (230425 bytes, image/jpeg)

OK s3://lakehouse/raw/visdrone/0000001/0000001.jpg (165707 bytes, image/jpeg)

OK s3://lakehouse/raw/visdrone/0000001/0000001-2.jpg (224567 bytes, image/jpeg)